

Smart Water Tank & Submersible Pump Automation

System Type: Dual-Node Wireless IoT Automated Fluid Management System

Microcontrollers: 2x ESP32 (Roof Node & Pump Node) **Communication:** ESP-NOW (Local Wireless) & Wi-Fi (Blynk 2.0 Cloud)

1. Executive Summary

This project upgrades a manual 1HP submersible water pump system into a fully automated, cloud-connected smart grid. It uses ultrasonic soundwaves to measure water levels in a concrete rooftop tank with millimetre accuracy. **The system automatically maintains water levels, prevents overflow, protects the pump from dry-running,** and provides real-time global monitoring via a mobile smartphone application.

2. Core System Architecture (How It Works)

The system is divided into three isolated layers:

1. **The Roof Node (The Sensor):** An ESP32 microcontroller paired with **a waterproof AJ-SR04M ultrasonic sensor sits above the concrete tank.** It bounces sound waves off the water's surface to calculate the exact percentage of water remaining. It wirelessly transmits this data every 2 seconds.
2. **The Pump Node (The Master Brain):** A second ESP32 sits downstairs near the KSB motor starter box. **It receives the wireless signals from the roof. It runs the logic to decide when to start and stop the pump,** and it connects to the home Wi-Fi to push data to the cloud.
3. **The Cloud Application (The UI):** The Blynk 2.0 platform receives data from the Pump Node, **displaying live gauges, historical water usage graphs, and an emergency manual override switch on the user's smartphone.**

3. Critical Safety Protocols (Electrical & Hardware)

Safety is the foundational priority of this build. The following strict protocols are engineered into the system:

- **Submersible Pump Protection (The "3-Second Rule"):** Submersible pumps require a massive surge of power from a Starting Capacitor to begin spinning. **The system uses a dedicated "Pulse Timer" algorithm.** Instead of holding the power on (which would explode the capacitor), **the microcontroller triggers a relay for exactly 3 seconds to mimic a human finger pressing the Green "Start" button, then releases it. A second relay is used to momentarily press the Red "Stop" button when the tank is full.**
- **Complete Electrical Isolation:** **The low-voltage microcontrollers (3.3V) are physically separated from the high-voltage mains (220V) using opto-isolated magnetic relays.**
- **Master Kill-Switch:** **A dedicated 6-Amp physical wall switch is installed on the main bathroom switchboard. Flipping this switch instantly cuts all power to the roof electronics during severe thunderstorms or maintenance.**
- **Weather & Environmental Proofing:** All exterior electronics are sealed inside an **IP65-rated PVC Waterproof Junction Box**. The wire entry points utilize gravity "drip loops" so rainwater cannot run into the enclosures. **The ultrasonic sensor is permanently bonded to the concrete lid using industrial-grade M-Seal epoxy putty to withstand extreme summer heat and prevent internal tank condensation from degrading the mount.**
- **Offline Redundancy:** If the home Wi-Fi router loses power or internet connection, the system automatically falls back to an offline ESP-NOW radio frequency (Channel 1). The pump will continue to fill the tank automatically without internet access.
- **Hard Overflow Shutoff:** Regardless of user input on the mobile app, if the water level reaches 100%, the system executes a mandatory hard-stop command to prevent roof flooding.

4. System Wiring & Schematics

Diagram A: Roof Node Assembly

```
[BATHROOM SWITCHBOARD]
|
(6A Master Kill Switch)
|
(Outdoor-Rated 220V Copper Wire)
|
v
=====
[IP65 WEATHERPROOF BOX - ROOF] (Zip-tied to tank stand)
|
|--> [5V Mobile Adapter] --> [ESP32 Microcontroller]
|                               |
|                               v
[CONCRETE TANK LID]           [AJ-SR04M Green Board]
|                               |
|-- (M-Seal Epoxy Mount)      |
|-- [Waterproof Ultrasonic Probe] <-----|
=====
```

Diagram B: Pump Node & KSB Starter Integration

```
[HOME WI-FI ROUTER] <----(Cloud Data)----> [BLYNK SMARTPHONE APP]
^
| (Wireless Connection)
v
=====
[INDOOR PVC BOX]
[ESP32 Microcontroller]
|
|--> (Digital Pin 19) --> [Relay Channel 1] (The "Start" Finger)
|
|--> (Digital Pin 18) --> [Relay Channel 2] (The "Stop" Finger)
=====
|                               |
v                               v
[KSB PUMP STARTER BOX]
|
|--> Relay 1 wired in parallel to the GREEN Start Button.
|   (ESP32 pulses this for 3 seconds to trigger the capacitor).
|
|--> Relay 2 wired in series with the RED Stop Button.
|   (ESP32 pulses this for 1 second to break the magnetic latch).
```

5. Implementation Phases

To ensure zero downtime for the household water supply, the installation is broken into two phases:

Phase 1: Roof Telemetry & Infrastructure

Run the 220V power line from the bathroom to the roof.

Mount the waterproof sensor to the concrete lid.

Install the IP65 junction box and activate the Roof ESP32.

Result: The household continues to use the pump manually, but gains real-time water level viewing on their smartphones.

Phase 2: Pump Automation Integration

Procure the required 2-Channel Relay Module.

Program the "3-Second Pulse" logic into the Pump ESP32.

Wire the relays into the KSB Starter Box.

Result: Full hands-free automation is achieved.

6. Electrical Specifications & Load Ratings

This section details the power consumption, operating voltages, and safety tolerances for both the heavy industrial load (the pump) and the low-voltage control circuits (the IoT nodes).

6.1. The Heavy Load: KSB Submersible Pump Motor

Operating Voltage: 220V - 240V AC (Single Phase, 50Hz)

Continuous Power Output: 1 Horsepower (Approx. 746 Watts)

Continuous Running Current: ~4.0 to 5.0 Amps

Starting Surge Current: 15 to 20 Amps

Engineering Note: This massive starting surge is handled strictly by the 120/150 MFD Starting Capacitor inside the KSB Starter Box. The IoT relays will never carry this heavy motor current directly; they only trigger the low-current contactor coil.

Overload Protection: Protected by a dedicated 10-Amp MCB integrated into the KSB Starter Box.

6.2. The Control Power Supplies: Itel Mobile Adapters (Model L25ZIIIE)

AC Input Rating: 100-240V AC, 50/60Hz, 0.2A

DC Output Rating: 5.0V DC at 500mA (0.5 Amps)

Maximum Power Delivery: 2.5 Watts

Safety Factor: These chargers utilize internal step-down transformers to physically and electrically isolate the 220V AC house mains from the 5V DC electronic components, preventing high-voltage shock hazards on the roof.

6.3. The Microcontrollers: ESP32 IoT Nodes

Operating Voltage: 5V DC (Input via Micro-USB) / 3.3V DC (Internal Logic)

Peak Current Draw: ~240mA (Brief spikes during wireless radio transmission)

Idle/Listening Current: ~40mA to 80mA

Total Power Consumption: < 1.5 Watts maximum

Safety Factor: The 500mA Itel charger provides more than double the required amperage (240mA) for the ESP32, ensuring the charger never overheats or strains under the load.

6.4. The Telemetry Sensor: AJ-SR04M Ultrasonic Board

Operating Voltage: 5V DC

Operating Current: ~30mA (Extremely low power)

Acoustic Emission: 40 KHz (Inaudible to humans and pets)

6.5. Wiring & Cabling Safety Margins

Roof Main Power Cable: 1.0 sq mm Outdoor-Rated Flexible Copper Wire.

Maximum Cable Capacity: Rated to safely carry continuous loads of **10 to 12 Amps (Approx. 2,200 to 2,600 Watts)** before heating occurs.

Actual Applied Load: The roof control system draws a maximum of **0.01 Amps (2.5 Watts)**.

Conclusion: The 220V wire running to the roof is operating at **less than 0.2% of its maximum thermal capacity**. It acts purely as a low-stress delivery line and poses zero risk of overheating, melting, or electrical fire.

7. Emergency Operating Procedure (EOP) / Manual Override

This section details how to bypass the automated system entirely in the event of a catastrophic hardware failure, Wi-Fi outage, or system glitch, returning the pump to 100% manual control.

Scenario: The tank is empty, the app is unresponsive, and the pump will not start automatically.

1.Kill the Smart System: Go to the 1st-floor bathroom and flip the dedicated "Smart Grid Master Switch" to the **OFF** position. This instantly cuts all power to the ESP32 microcontrollers and the relay modules.

2.Return to Manual Mode: Walk to the KSB Starter Box.

3.Start the Pump: Press and hold the Green "Start" button for 3 seconds, then release (Standard operating procedure).

4.Stop the Pump: When the tank overflows or is full, press the Red "Stop" button.

Engineering Note: Because the smart relays are wired "in parallel" to the physical buttons, turning off the smart system makes the buttons behave exactly as they did before the installation. The house is never locked out of its own water supply.

8. Maintenance & Troubleshooting Protocol

The system is designed to run maintenance-free for years, but the following standard operating procedures should be observed to ensure longevity:

- **Biannual Sensor Cleaning (Every 6 Months):** Dust, spider webs, or hard water condensation can eventually build up on the face of the AJ-SR04M ultrasonic probe. Once every six months, wipe the face of the blue sensor probe with a dry microfiber cloth to ensure the soundwaves remain unobstructed.
- **Wi-Fi Network Changes:** If the household router is replaced or the Wi-Fi password is changed, the Pump Node will be unable to reach the Blynk Cloud.
 - *System Response:* The Pump Node will automatically fall back to "Offline Auto-Mode" (ESP-NOW Channel 1). It will continue to fill the tank automatically, but the mobile app will show the device as "Offline."
 - *Resolution:* The Pump ESP32 must be plugged into a computer via USB to flash the new Wi-Fi credentials into the code.
- **Power Outages:** Upon restoration of house power, the system requires exactly 10 seconds to reboot, reconnect to the cloud, and measure the water level before resuming standard automated logic.

9. Bill of Materials (BOM) & Budget Estimate

This project utilizes highly cost-effective, readily available micro-electronics, making repairs or part replacements incredibly cheap compared to commercial smart-home solutions.

Hardware & Electronics:

2x ESP32 NodeMCU Microcontrollers

1x AJ-SR04M Waterproof Ultrasonic Sensor

1x 5V 2-Channel Opto-isolated Relay Module

2x 5V 500mA Itel Mobile Adapters (Power Supply)

Infrastructure & Casing:

2x IP65 Weatherproof PVC Junction Boxes (8x6 inch)

~15 Meters of 1.0 sq mm Outdoor-Rated 2-Core Copper Wire

1x 6-Amp Physical Wall Switch & 2x 3-Pin Sockets

M-Seal Epoxy Putty & Heavy-Duty Double-Sided Tape

Total Estimated Hardware Cost: ₹1,500 – ₹2,000 INR

Commercial Equivalent Cost: ₹8,000+ INR

Cost Savings: ~75%